Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	05 April 2026
Team ID	SWTID-2026-5577
Project Name	Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking
Maximum Marks	4 Marks

Technical Architecture

1. **Data Input Layer**: External Excel/CSV files containing employee and organizational data.
2. **Staging Layer**: The ServiceNow Import Set table (Import Employee), which acts as a temporary buffer to validate data before committing to the system.
3. **Logic Layer**: Transform Maps and Mapping Assist tools that execute the movement and relationship linking (User to Manager/Department).
4. **Data Storage Layer**: The permanent ServiceNow Target Tables (specifically the sys_user table).
5. **Interface Layer**: The Incident Form layout, enhanced with Dot Walking to pull related data dynamically for end-users.

Table 1: Technology Stack Description

S.No	Component	Description	Technology Used
1.	User Interface	How user interacts with application	ServiceNow UI (Portal & Native Forms) + HTML/CSS (for customizations)
2.	Application Logic-1	Data import from external sources	ServiceNow Data Sources (CSV, Excel, Database connectors)
3.	Application Logic-2	Staging and transformation logic	Import Sets & Transform Maps (ServiceNow native ETL tools)
4.	Application Logic-3	Relationship mapping & data access	Reference Fields + Dot Walking (GlideRecord API)
5.	Database	Primary data storage	MySQL (ServiceNow underlying DB) + ServiceNow CMDB
6.	Cloud Database	Cloud-based database service	ServiceNow Cloud Database (hosted on AWS/GCP)
7.	File Storage	File storage requirements	ServiceNow Attachment Storage (Cloud-based) / Local filesystem for source files
8.	External API-1	Purpose of External API used	ServiceNow REST API (for external data push/pull if needed)
9.	External API-2	Purpose of External API used	Google Sheets API (optional, for direct cloud import)

S.No	Component	Description	Technology Used
10.	Machine Learning	Purpose of ML Model	Not applicable (rule-based transformation used; ML can be added for duplicate detection)
11.	Infrastructure	Application deployment	ServiceNow Cloud Platform (Multi-instance, SaaS)

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology Used
1.	Open-Source Frameworks	List of open-source frameworks used	None directly (ServiceNow is proprietary). For custom scripts: JavaScript (ECMAScript 5/6)
2.	Security Implementations	Security and access controls implemented	ServiceNow ACLs (Access Control Lists), RBAC (Role-Based Access Control), HTTPS/TLS 1.2+, OAuth 2.0 for APIs, and Instance-level firewalls.
3.	Scalable Architecture	Justification of scalability	Multi-tier (3-Tier): Presentation (UI) → Application (Import Sets) → Data (Database). ServiceNow auto-scales horizontally across cloud nodes.
4.	Availability	Justification of availability	99.99% SLA by ServiceNow. Multi-data center replication with automatic failover and Load balancers to distribute traffic.

S.No	Characteristics	Description	Technology Used
5.	Performance	Design consideration for performance	Import Set batch processing for high-volume records, Transform Map commit batching, and Dot Walking to avoid N+1 queries.